

Unit 5 Guided Notes — The Pythagorean Theorem

$a^2 + b^2 = c^2$. Add the legs to find the hypotenuse; subtract to find a leg.

🎯 By the end of this unit, I can...

- Use the Pythagorean Theorem to find a missing side of a right triangle
- Apply its converse to test whether a triangle is right
- Solve real-world and coordinate-distance problems

Standards: NC.8.G.6 · NC.8.G.7 · NC.8.G.8

📄 Blank Worksheet (PDF)

✅ Answer Key (PDF)

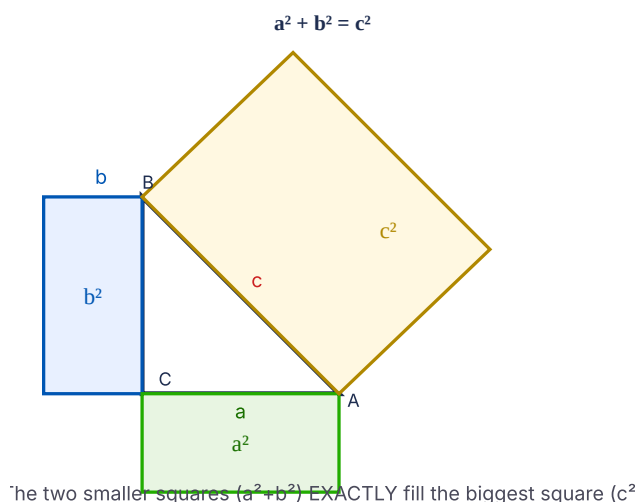
📁 Official PDF Packet

This packet has three parts: **Part 1 — Guided Notes** (fill in the blanks as you learn), **Part 2 — Practice Problems**, and the **Answer Key** built right in. $a^2 + b^2 = c^2$ — add the legs to find the hypotenuse; subtract to find a leg.

1 The Pythagorean Theorem

🔑 **THE THEOREM:** In a RIGHT triangle, the sum of the squares of the two LEGS equals the square of the HYPOTENUSE: $a^2 + b^2 = c^2$.

a, b = legs (shorter sides) · c = hypotenuse (longest side, across from the right angle)



⚠️ **IMPORTANT:** The Pythagorean Theorem ONLY works on RIGHT triangles (triangles with a 90° angle)!

Complete each blank on your printed copy, then click to check.

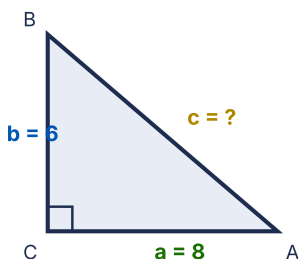
The two shorter sides of a right triangle are called the **LEGS** (labeled a and b). The longest side, directly across from the right angle, is the **HYPOTENUSE** (labeled c).

The theorem written as an equation: $a^2 + b^2 = c^2$. It applies only to **RIGHT** triangles.

2 Finding the Hypotenuse

WHEN TO USE: When you know BOTH LEGS and need to find the HYPOTENUSE. Formula:
 $c^2 = a^2 + b^2 \rightarrow c = \sqrt{a^2 + b^2}$

FIND THE HYPOTENUSE



Example: Legs are 6 and 8. Find the hypotenuse.

Step 1 – Write formula: $a^2 + b^2 = c^2$

Step 2 – Substitute: $6^2 + 8^2 = c^2$

Step 3 – Square: $36 + 64 = c^2$

Step 4 – Add: $100 = c^2$

✓ **Step 5 – Square root:** $c = \sqrt{100} = 10$

Practice — Find the Hypotenuse (c)

Legs 3 and 4: $3^2 + 4^2 = c^2 \rightarrow c = ?$

Answer: 5

Legs 6 and 8: $6^2 + 8^2 = c^2 \rightarrow c = ?$

Answer: 10

Legs 5 and 12: $5^2 + 12^2 = c^2 \rightarrow c = ?$

13

Answer: 13

Legs 8 and 15: $8^2 + 15^2 = c^2 \rightarrow c = ?$

17

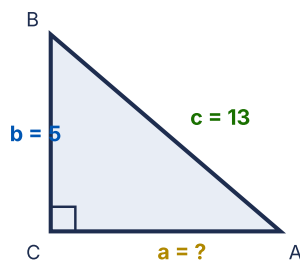
Answer: 17

3 Finding a Missing Leg

🔑 WHEN TO USE: When you know ONE LEG and the HYPOTENUSE. Formula: $a^2 = c^2 - b^2 \rightarrow a = \sqrt{c^2 - b^2}$

⚠️ REMEMBER: Finding a LEG = SUBTRACT from the hypotenuse squared. Finding the HYPOTENUSE = ADD the two legs squared.

FIND THE MISSING LEG



Example: One leg is 5, hypotenuse is 13. Find the other leg.

Step 1 – Write formula: $a^2 + b^2 = c^2$

Step 2 – Substitute: $5^2 + b^2 = 13^2$

Step 3 – Square: $25 + b^2 = 169$

Step 4 – Subtract 25 (finding a LEG = SUBTRACT): $b^2 = 144$

✓ **Step 5 – Square root:** $b = \sqrt{144} = 12$

ERROR ANALYSIS A student finds a missing leg when $c = 13$, $b = 5$ by writing

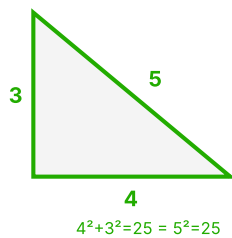
$a^2 = 13^2 + 5^2 = 194$. What's wrong?

To find a leg you SUBTRACT: $a^2 = c^2 - b^2 = 169 - 25 = 144 \rightarrow a = 12$. Adding is only for the hypotenuse.

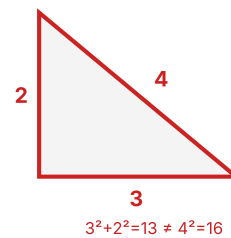
4 Converse of the Pythagorean Theorem

THE CONVERSE: Given three side lengths $a \leq b \leq c$, test: does $a^2 + b^2 = c^2$? If YES $\rightarrow$ right triangle. If NO $\rightarrow$ not a right triangle. Always test the two SMALLER sides against the LARGEST.

3, 4, 5
RIGHT triangle ✓

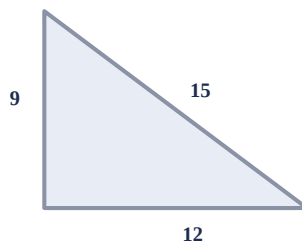


2, 3, 4
NOT a right triangle ✗



Example: Sides 9, 12, 15. Is it a right triangle?

9, 12, 15



Step 1 – Order the sides: $9 \leq 12 \leq 15$, so the LARGEST is $c = 15$

Step 2 – Write the test: does $a^2 + b^2 = c^2$?

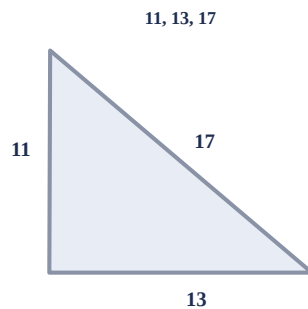
Step 3 – Substitute the two SMALLER sides: $9^2 + 12^2 = ? 15^2$

Step 4 – Square each one: $81 + 144 = ? 225$

Step 5 – Add the left side: $225 = ? 225$

✓ **Step 6 – Compare and conclude:** The two sides are EQUAL, so YES – it is a right triangle.

⚠ **Your eyes cannot do this job.** Every triangle below is drawn to true scale. Look at **11, 13, 17**: it is *not* a right triangle, because $11^2 + 13^2 = 121 + 169 = \mathbf{290}$ while $17^2 = \mathbf{289}$. It misses by one. Drawn accurately it sits two-tenths of a degree off square — nobody alive could spot that. This is exactly why the converse exists: **you must run the arithmetic, every time.**



Quick Checks

MC Sides 5, 12, 13 — right triangle?

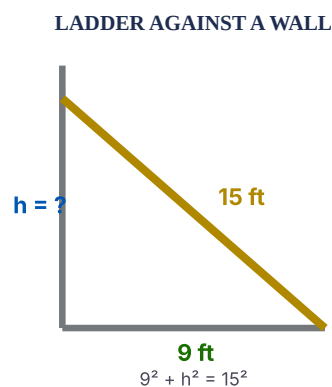
✓ $5^2 + 12^2 = 25 + 144 = 169$; $13^2 = 169$; $169 = 169 \rightarrow$ YES, right triangle.

MC Sides 6, 7, 10 — right triangle?

✓ $6^2 + 7^2 = 36 + 49 = 85$; $10^2 = 100$; $85 \neq 100 \rightarrow$ NO, not a right triangle.

5 Word Problems

📌 **STRATEGY:** (1) Read carefully (2) Draw & label a diagram (3) Identify legs vs hypotenuse (4) Set up $a^2 + b^2 = c^2$ (5) Solve step by step.



Ladder Against a Wall: A ladder 15 ft long leans against a wall. The base is 9 ft from the wall. How high does it reach?

Setup: $9^2 + h^2 = 15^2$

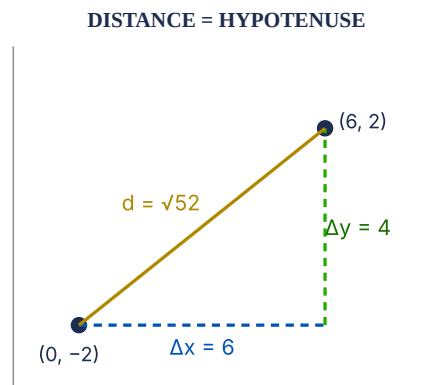
Square & subtract: $81 + h^2 = 225 \rightarrow h^2 = 144$

✓ **Square root:** $h = \sqrt{144} = 12 \text{ ft}$

6 Distance on the Coordinate Plane

📐 **HOW IT WORKS:** Connect the two points to form a right triangle. $\Delta x = |x_2 - x_1|$ and $\Delta y = |y_2 - y_1|$. Then use $a^2 + b^2 = c^2$ where $a = \Delta x$, $b = \Delta y$, and $c = \text{distance}$.

⚠️ **NO DISTANCE FORMULA NEEDED!** You already know how to do this — just use the Pythagorean Theorem!



Example: Distance from $(0, -2)$ to $(6, 2)$

Δx : $|6 - 0| = 6$

Δy : $|2 - (-2)| = 4$

Setup: $\Delta x^2 + \Delta y^2 = d^2 \rightarrow 36 + 16 = 52$

✓ **Answer:** $d = \sqrt{52} \approx 7.21$

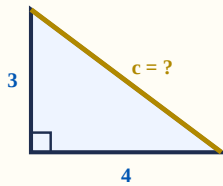
⚠️ **WATCH YOUR SIGNS:** When subtracting negative coordinates, subtracting a negative means ADDING! Example: $5 - (-3) = 5 + 3 = 8$, NOT $5 - 3 = 2$.

Part 2 — Practice Problems

 Show all work including the formula setup.

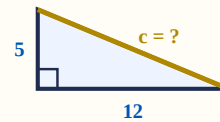
A. Find the Hypotenuse

1) Legs 3, 4. $c = ?$



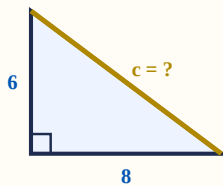
Answer: 5

2) Legs 5, 12. $c = ?$



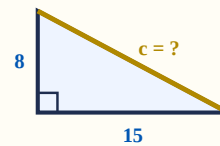
Answer: 13

3) Legs 6, 8. $c = ?$



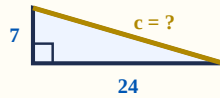
Answer: 10

4) Legs 8, 15. $c = ?$



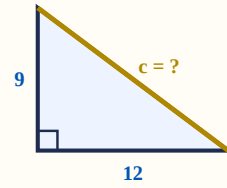
Answer: 17

5) Legs 7, 24. $c = ?$



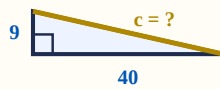
Answer: 25

6) Legs 9, 12. $c = ?$



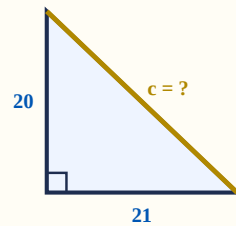
Answer: 15

7) Legs 9, 40. $c = ?$



Answer: 41

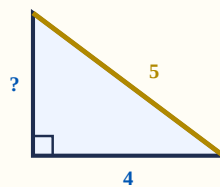
8) Legs 20, 21. $c = ?$



Answer: 29

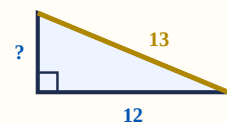
B. Find the Missing Leg

1) Known leg 4, hyp 5. Leg = ?



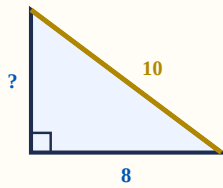
Answer: 3

2) Known leg 12, hyp 13. Leg = ?



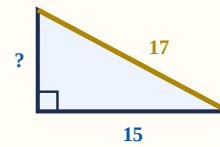
Answer: 5

3) Known leg 8, hyp 10. Leg = ?



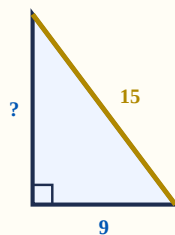
Answer: 6

4) Known leg 15, hyp 17. Leg = ?



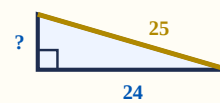
Answer: 8

5) Known leg 9, hyp 15. Leg = ?



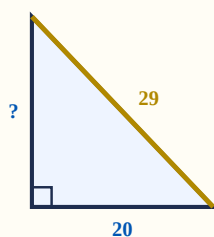
Answer: 12

6) Known leg 24, hyp 25. Leg = ?



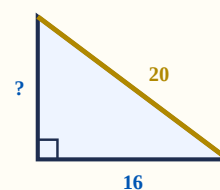
Answer: 7

7) Known leg 20, hyp 29. Leg = ?



Answer: 21

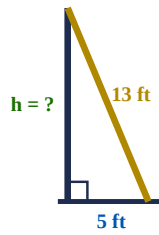
8) Known leg 16, hyp 20. Leg = ?



Answer: 12

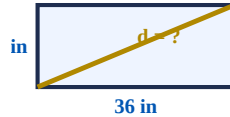
C. Word Problems

1. A 13-foot ladder leans against a wall. The base is 5 feet from the wall. How high does it reach?



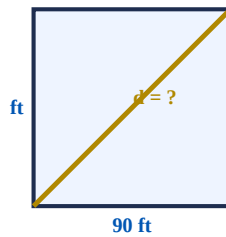
$$h^2 + 5^2 = 13^2 \rightarrow h^2 = 169 - 25 = 144 \rightarrow h = 12 \text{ ft.}$$

2. A rectangular TV is 36 inches wide and 15 inches tall. What is the diagonal length?



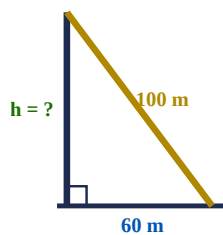
$$d^2 = 36^2 + 15^2 = 1296 + 225 = 1521 \rightarrow d = 39 \text{ in.}$$

3. A baseball diamond is a square with 90-ft sides. Distance from home to second base?



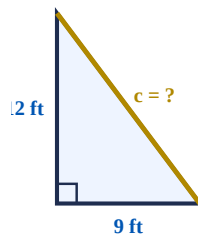
$$d^2 = 90^2 + 90^2 = 16200 \rightarrow d \approx 127.3 \text{ ft.}$$

4. A kite string is 100 m long. The kite is directly above a point 60 m away. How high is the kite?



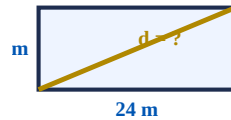
$$h^2 + 60^2 = 100^2 \rightarrow h^2 = 10000 - 3600 = 6400 \rightarrow h = 80 \text{ m.}$$

5. A support beam covers 9 ft horizontal and 12 ft vertical. How long is the beam?



$$c^2 = 9^2 + 12^2 = 81 + 144 = 225 \rightarrow c = 15 \text{ ft.}$$

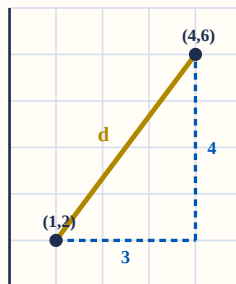
6. A pool is 24 m long and 10 m wide. A swimmer swims diagonally. How far?



$$d^2 = 24^2 + 10^2 = 576 + 100 = 676 \rightarrow d = 26 \text{ m.}$$

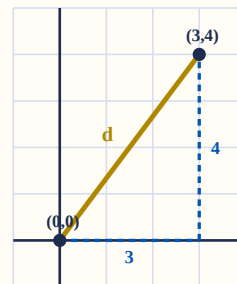
D. Distance on the Coordinate Plane

1) $(1,2) \rightarrow (4,6)$. $d = ?$



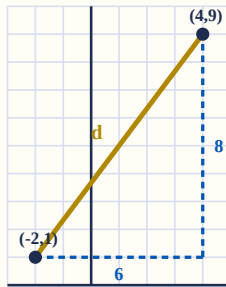
Answer: 5

2) $(0,0) \rightarrow (3,4)$. $d = ?$



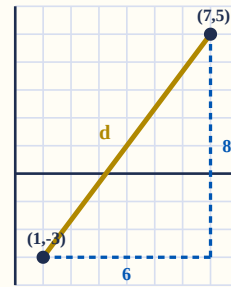
Answer: 5

3) $(-2,1) \rightarrow (4,9)$. $d = ?$



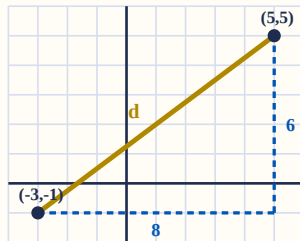
Answer: 10

4) $(1,-3) \rightarrow (7,5)$. $d = ?$



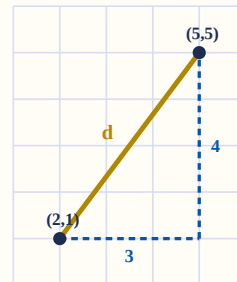
Answer: 10

5) $(-3,-1) \rightarrow (5,5)$. $d = ?$



Answer: 10

6) $(2,1) \rightarrow (5,5)$. $d = ?$

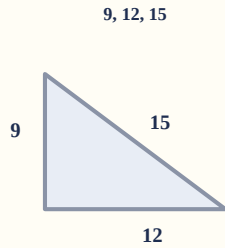


Answer: 5

E. The Converse — Is It a Right Triangle?

Test the two **smaller** sides against the **largest**. Each figure is drawn to true scale, so do not trust the picture — two of these miss by a hair.

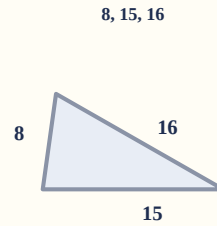
1) Sides 9, 12, 15 — right triangle? (type yes or no)



yes / no

Answer: Yes — $9^2 + 12^2 = 225$, and $15^2 = 225$, so $225 = 225$

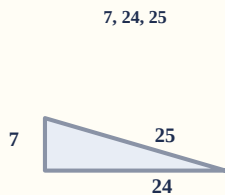
2) Sides 8, 15, 16 — right triangle? (type yes or no)



yes / no

Answer: No — $8^2 + 15^2 = 289$, and $16^2 = 256$, so $289 \neq 256$

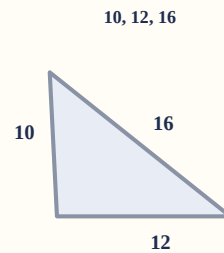
3) Sides 7, 24, 25 — right triangle? (type yes or no)



yes / no

Answer: Yes — $7^2 + 24^2 = 625$, and $25^2 = 625$, so $625 = 625$

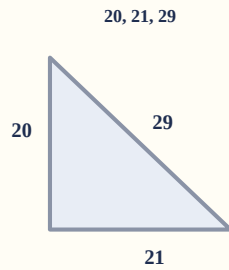
4) Sides 10, 12, 16 — right triangle? (type yes or no)



yes / no

Answer: No — $10^2 + 12^2 = 244$, and $16^2 = 256$, so $244 \neq 256$

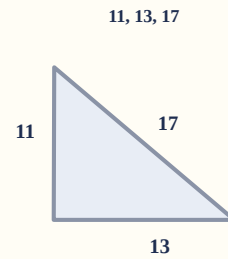
5) Sides 20, 21, 29 — right triangle? (type yes or no)



yes / no

Answer: Yes — $20^2 + 21^2 = 841$, and $29^2 = 841$, so $841 = 841$

6) Sides 11, 13, 17 — right triangle? (type yes or no)



yes / no

Answer: No — $11^2 + 13^2 = 290$, and $17^2 = 289$, so $290 \neq 289$

← Back to Unit 5